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M3.F1 Chatbot Lab
COSC 524
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Reflections

What did your exploration of architecture structure reveal?

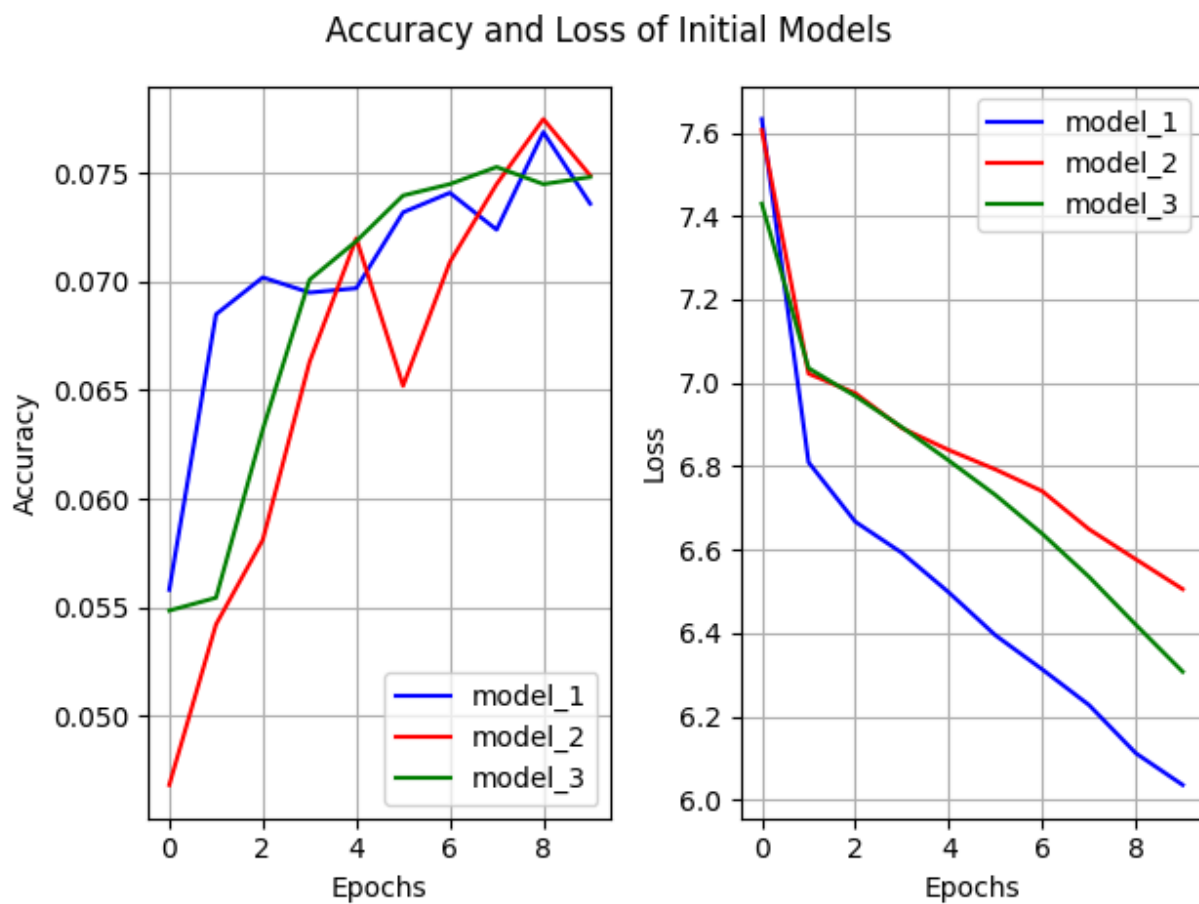


Figure 1: Initial model accuracy and loss. Model 1: 1 layer, 50 nodes, 10 epochs; Model 2: 5 layers, 50 nodes, 10 epochs; Model 3: 5 layers, 100 nodes, 10 epochs

One major thing I learned during testing my model is that more layers doesn't always equal a better output! In figure 1, I trained my model using a small amount of my data for time's sake. 10 epochs is also incredibly short, so I wasn't expecting great results out of these (hence the <10% accuracy). I was surprised that the models with 5 layers

weren't performing that much better than the model with the single layer. For that reason, I settled on 2 layers for my final architecture. I also chose to increase the number of nodes in the final architecture.

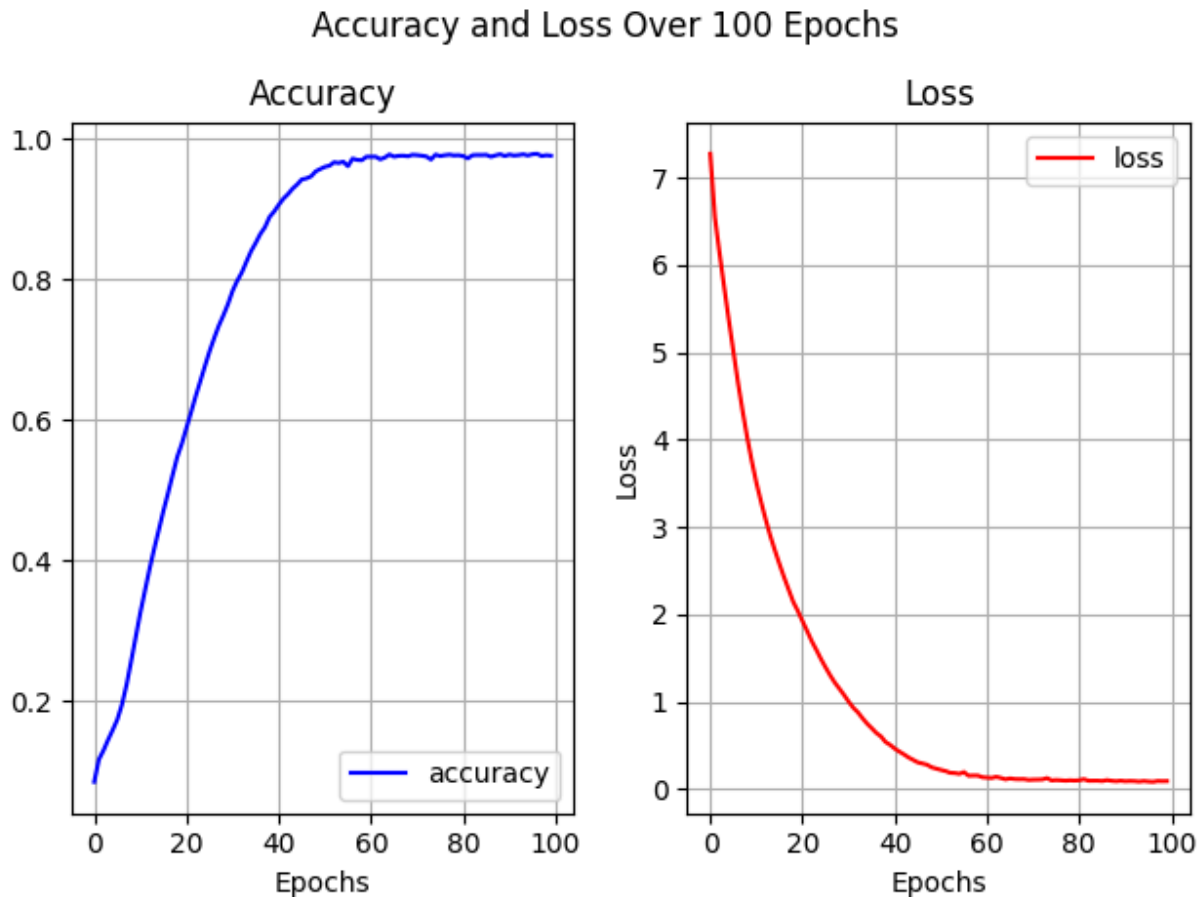


Figure 2: Accuracy and Loss for final model. 100 epochs, 2 layers, 256 nodes

The final model showed improvements much more quickly than the previous models, almost reaching 0.2 at 10 epochs, where the previous ones did not see an accuracy higher than 0.08 at 10 epochs. The model begins to plateau around epoch 60, so 100 epochs may be overshooting it. The output of the model did not make much sense grammatically, but there were like words together (such as *cloud*, *cumulonimbus*, and *front*). Because I did not set aside a subset of the corpus to use as validation data, it is hard to tell if the model is experiencing overfitting or underfitting, so I plan to retrain it with validation data so that I can watch the trends of *val_loss* and *val_accuracy*. I will also reduce the number of epochs and perhaps the number of nodes.

What impact did increasing/decreasing the training epochs have on the time it took to train the model and the quality of the generated text?

The model took significantly longer to train the more epochs that were used, but the model also had the time to improve significantly.

What impact did using smaller- and larger-sized documents have on the time it took to train the model and the quality of the generated text?

Using only an incredibly small portion of the total corpus produced incredibly quick training times (< 10 minutes), but the output was completely random. The generated output began to make more sense as the size of the corpus increased, but it also took the model a significantly longer time to train. The final model, which included the entire corpus, took around 12 hours to train.

Sample Text Output

The research shows to rapid characteristics and the transport of access of tidal origin and through the uv may gas shelves during an object up to the source of stable planets with high resolution a cloud pressure also may be outgassed the chemical composition of cloud droplets has been high amounts of clouds in the frontal zone' or by cumulonimbus ice or hot life and high warming slows

Future Improvements

I believe that this model could be improved by potentially having fewer nodes and fewer epochs. I would also want to look into how to include punctuation in the model's prediction so that the output looks more like standard writing. I would also want to separate validation data so that I can see if the model is experiencing overfitting or underfitting, and address it as appropriate.